# The conjectures in *A Bayesian Algorithm for Optimal Dating Expenditure*: proofs, counterexamples and corrected statements

Companion note

30 October 2026

**Abstract**

The proposal states five conjectures (Conjectures 1–5) and leaves one question open: whether the value of information can be negative once refusals sting (Section 4.4). This note tests each conjecture in the core model (M2), where the value after the date is truncated after one more date.

- Conjectures 1 and 5 are false as stated. Conjectures 2 and 3 are false as stated for decisions on the day. Each is refuted by an explicit counterexample, and each has a true core, which is proved under conditions stated exactly.
- Conjectures 2 and 3 are posed in M2+. The note settles only their core-model analogues; the M2+ versions remain open.
- Conjecture 4 is conditional on concavity, which the note shows fails whenever entry prices bind. Its threshold and spend claims are proved, without the premise in the core model and under it at the last step of M2+. No counterexample satisfying the premise is given. Its claim about cheaper opening activities does not follow from the shadow-value logic, and fails once the premise is dropped.

The open question is answered for signals about compatibility or about willingness alone. Information about compatibility always has non-negative value. So does information about willingness from monotone signals, such as the guide's cues. A non-monotone signal about willingness can have strictly negative value under all three models of the sting. Every number in this note is reproduced by `verify.py`, which also runs randomised checks of each theorem.

## Summary

| Claim | Verdict | What is proved |
|---|---|---|
| Conjecture 1 (threshold rules) | False in general (Prop. 8) | Threshold rules hold under a single condition (M) (Theorem 6). The condition is automatic when the after-date choice does not depend on what the date reveals. With two types it is implied by $v_j(C_H - C_L) \geq \tilde{\kappa}\,\lvert\psi(C_H) - \psi(C_L)\rvert$ (Corollary 7). |
| Conjecture 2 (outside option; posed in M2+) | Core-model analogue: true for Phase C, false on the day in general. M2+ version open | Raising $U$ shrinks the set of beliefs at which the agent asks for another date. On the day, raising $U$ makes continuing *less* attractive at beliefs where the agent would cut contact after the date, and *more* attractive where they would ask for another date (Lemma 10, Theorem 11). |
| Conjecture 3 (relationship value; posed in M2+) | First part, core-model analogue: as for Conjecture 2, with signs reversed. M2+ version open. Second part: false when uncertainty is compared across beliefs | Raising $R$ has the mirror-image effect of raising $U$ (Theorem 13). The investment value of a step peaks where the after-date choice is closest to indifference, can be zero at maximal uncertainty, and vanishes as beliefs settle (Proposition 15). |
| Conjecture 4 (tight budgets) | Premise fails near entry prices; no counterexample under it | Concavity gives the shadow-value claim by definition, but entry prices make $V$ non-concave in $B$. Thresholds rise as the budget falls: in the core model without concavity, and at the last step of M2+ when $V_{\text{end}}$ is concave in $B$. "Cheaper openings" does not follow from the shadow-value logic, and fails without the premise (Proposition 16). |
| Conjecture 5 (spending to learn) | Premise false; "falls back as beliefs settle" true; "peaks under uncertainty" false | EVSI need not be largest when beliefs are least settled. Optimal spend is never below the Proposition 1 level, under all three sting models, and converges to it as beliefs settle. The premium can be zero at maximal uncertainty and positive elsewhere. The caveat about stings is moot for this information (Proposition 21). |
| Open question, Sec. 4.4 | Answered for signals about $C$ alone or $q$ alone | EVSI $\geq 0$ for every signal about $C$, and for every MLRP signal about $q$, under D1, D2 and KR. A non-MLRP signal about $q$ can have negative EVSI under D1, D2 and KR (Theorem 24, Examples 25 and 26). |

# 1 Setting

Notation is that of the proposal. Throughout, $b$ denotes the agent's belief, $\mathbb{E}_b$ expectation under it, $p = \varphi_j \mathbb{E}_b[q]$ the probability of a yes at step $j$, and

$$\hat{p}(p) \;=\; p - \kappa_m(p), \qquad m \in \{\text{D1}, \text{D2}, \text{KR}\},$$

the *effective continuation weight*: $\kappa_{\text{D1}} = \eta(1-p)$, $\kappa_{\text{D2}} = \eta p(1-p)$, $\kappa_{\text{KR}} = \eta(\Lambda - 1)p(1-p)$.

**Standing assumption S1** (Core model)**.** Assumptions A1–A12 hold, including independent priors (A5). The value after the date is truncated after one more date, as in (19)–(21), with $h(c, q_D) = \bar{h}\, c\, q_D$.

**Standing assumption S2** (Ratings)**.** After step $j$ the agent observes a rating $y$ whose likelihood $P_j(y \mid c)$ has the monotone likelihood ratio property (MLRP) in $c$, and updates by Bayes' rule. The leading case is $y \sim \text{Bin}(m_j, C)$ with integer $m_j$. Beta beliefs then stay Beta: $\text{Beta}(\alpha, \beta) \mapsto \text{Beta}(\alpha + y, \beta + m_j - y)$.

*Remark* 1 (The fractional likelihood). The fractional Binomial (2) is a tempered quasi-likelihood. It does not by itself define a distribution for the rating, so expected posterior moments need not equal prior moments. Under it, $\mathbb{E}[\mathbb{E}_{b'} f(C) \mid b] = \mathbb{E}_b f(C)$ holds for $f(c) = c$ when $\mathbb{E}[\tilde{y} \mid C] = m_a C$, but not for nonlinear $f$. The results below that use the martingale property for nonlinear functions of $C$ therefore hold exactly with integer $m_a$ and exact Bayes, and approximately with fractional $m_a$. The results that use only monotonicity need, in addition, a generative model for the rating. For example, take $y \sim \mathrm{Bin}(10, C)$ with the tempered update (3). The posterior is then monotone in $y$ and in the prior, and the predictive distribution of $y$ is stochastically increasing in the belief.

**Standing assumption S3** (No sting without a stake)**.** The expected feeling of an answer is $\kappa_m(p)\,\Delta^+$ with $\Delta^+ = \max\{\Delta, 0\}$. The proposal assumes $\Delta > 0$ throughout; this extends it to $\Delta \le 0$ without turning a refusal into a gain.

**Standing assumption S4** (Willingness to meet again)**.** Where stated, $q_D$ is known, so $P_D = q_D$. This is part of the hypothesis of Conjecture 1. It is used elsewhere only to make the after-date value linear-plus-option in the belief about $C$ (Lemma 2); the same results hold if $q_D$ is uncertain and $\eta = 0$ in Phase C.

Write $\tilde{\kappa} = (\hat{p}(q_D))^+ = (q_D - \kappa_m(q_D))^+$, and define

$$a(c) = h(c)\,c\,R + \big(1 - h(c)\big)U,$$
$$\psi(c) = \big(1 - h(c)\big)\big[\delta_D\big(h(c)\,c\,R + (1 - h(c))U\big) - U\big], \qquad h(c) = \bar{h}\,q_D\,c,$$

and the *Phase-C index* $\Psi(b) = \mathbb{E}_b[\psi(C)]$.

**Lemma 2** (After-date value)**.** *Under S1, S3 and S4,*

$$V_{\mathrm{end}}(b) \;=\; \mathbb{E}_b[a(C)] \;+\; \tilde{\kappa}\;(\Psi(b))^+ .$$

*After a date that did not become a relationship, the agent asks for another date if and only if* $\tilde{\kappa} > 0$ *and* $\Psi(b) > 0$.

*Proof.* With $q_D$ known, a yes to another date says nothing about $C$, so in (20) $b_{\mathrm{yes}} = b$ and the stake is $\Delta_D = V_D(b_{\mathrm{not}}) - U$. If $\Delta_D \ge 0$, asking is worth $U + \hat{p}(q_D)\Delta_D$, so $V_C(b_{\mathrm{not}}) = U + \tilde{\kappa}\Delta_D$. If $\Delta_D < 0$, asking is worth $U + q_D\Delta_D < U$, so $V_C = U$. Hence $V_C(b_{\mathrm{not}}) = U + \tilde{\kappa}\,\Delta_D^+$. The event "no relationship" has likelihood $1 - h(C)$, so $\mathbb{E}_{b_{\mathrm{not}}}[f(C)]\,(1 - \mathbb{E}_b h) = \mathbb{E}_b[(1 - h(C))f(C)]$ for every $f$. Applying this to (21) gives $(1 - \mathbb{E}_b h)(V_D(b_{\mathrm{not}}) - U) = \mathbb{E}_b\psi = \Psi(b)$. Since $1 - \mathbb{E}_b h > 0$, $(1 - \mathbb{E}_b h)\Delta_D^+ = \Psi(b)^+$. Substituting into (19):

$$V_{\mathrm{end}}(b) = \mathbb{E}_b[hCR] + (1 - \mathbb{E}_b h)U + \tilde{\kappa}\,\Psi(b)^+ = \mathbb{E}_b[a] + \tilde{\kappa}\,\Psi(b)^+. \qquad \square$$

So $V_{\mathrm{end}}$ is linear in $b$ plus a convex option term. The option has a kink exactly where the after-date choice switches between cutting contact and asking for another date.

**Lemma 3** (A refusal carries no news; effective continuation weight)**.** *Under S1 and S3, the loss from a no is* $L = 0$*, so* $\ell = 0$ *and* $\Delta = G$*, at every step of every date. Writing*

$$G_j(b, B) = \sup_{s \in S_j(B)} \Big\{ W_j(b; s) - c_j^{\mathrm{time}} + \mathbb{E}\big[V_{j+1}(b', B - \sigma(k_j + s)) \mid b_{\mathrm{yes}}\big] \Big\} - V_{\mathrm{end}}(b)$$

*(with* $G_j = -\infty$ *if* $S_j(B) = \emptyset$*), the Bellman equation (8) is equivalent to*

$$V_j(b, B) = V_{\mathrm{end}}(b) + \hat{p}_j^{+}\,G_j(b, B)^+, \qquad \hat{p}_j = \hat{p}(\varphi_j \mathbb{E}_b q). \tag{1}$$

*The agent asks if and only if* $\hat{p}_j > 0$ *and* $G_j \ge 0$*, and the optimal venue maximises the term in braces.*

*Proof.* By Lemma 2, or directly from (19)–(21), $V_{\text{end}}$ depends on $b$ only through its marginal on $(C, q_D)$. By A4 and A5 an answer is informative only about $q$, which is independent of $(C, q_D)$. Hence $V_{\text{end}}(b_{\text{no}}) = V_{\text{end}}(b_{\text{yes}}) = V_{\text{end}}(b)$, so $L = 0$. At spend $s$, let $G(s)$ be the term in braces minus $V_{\text{end}}(b)$. Asking is worth $V_{\text{end}}(b) + pG(s) - \kappa(p)G(s)^+$. This is $V_{\text{end}} + \hat{p}\,G(s)$ when $G(s) \geq 0$, and is below $V_{\text{end}}$ when $G(s) < 0$. Maximising over $s$ and comparing with ending gives (1). □

The proposal's Remark after Proposition 3 already observes $\ell = 0$ under independence with the truncated $V_{\text{end}}$. Lemma 3 records it at every step, together with the resulting form of the value. The lemma depends on the truncation: with the full recursion of M2+, a refusal lowers $\mathbb{E}[q]$ and hence $V_{\text{end}}$, so $\ell > 0$, as the proposal notes.

Since $\hat{p}^+ \in [0, 1]$, (1) can be rewritten as

$$V_j = \max\Big\{V_{\text{end}}(b),\ \hat{p}_j^+\big(G_j + V_{\text{end}}(b)\big) + (1 - \hat{p}_j^+)V_{\text{end}}(b)\Big\}.$$

This is the Bellman equation of a *fictitious optimal stopping problem* with the same value function. After each ask the date continues with probability $\hat{p}^+$ and pays the step's flow. With probability $1 - \hat{p}^+$ it ends at the current belief about $C$, and when the date ends the agent receives $V_{\text{end}}$ at the terminal belief. Call this fictitious measure $\tilde{\mathbb{P}}$, with expectation $\tilde{\mathbb{E}}$. Because $\hat{p}$ does not depend on $C$, and ratings are Bayesian, the belief about $C$ is a $\tilde{\mathbb{P}}$-martingale. So for every policy $\pi$ and stopping time $\tau \leq J + 1$,

$$\tilde{\mathbb{E}}^{\pi}\big[\mathbb{E}_{b_\tau} f(C)\big] = \mathbb{E}_b f(C) \qquad \text{for every bounded } f. \tag{2}$$

This is the tool behind most of what follows.

The two problems differ only in the payoff of asking when $G_j < 0$. Asking is then worth $V_{\text{end}} + p\,G_j$ in the real problem and $V_{\text{end}} + \hat{p}\,G_j$ in the fictitious one. Neither choice is optimal in either problem, and the two expressions have the same sign when $\hat{p} > 0$, so values and decisions coincide.

*Remark* 4 (Closed form of Proposition 1). With $\bar{s}_j = k_j$ and $x = \mathbb{E}[C]v_j$, the net value of a step at the optimal spend is

$$W_j = x \ln\frac{x}{\sigma\lambda k_j} \quad \text{if } x \geq \sigma\lambda k_j, \qquad W_j = x - \sigma\lambda k_j \quad \text{otherwise.}$$

Discretionary spend is therefore positive exactly when the step is worth taking at its entry price.

## 2 Conjecture 1: threshold rules

Conjecture 1 concerns the one-dimensional case, with $q$ and $q_D$ known. Then $p_j = \varphi_j q$ and $\hat{p}_j$ are constants, and the agent asks at step $j$ iff $\hat{p}_j > 0$ and $G_j \geq 0$ (Lemma 3). At each step the belief lies in a totally ordered family $F_j$:

- two types, $b = \pi = P(C = C_H)$; or
- Beta, $b = \text{Beta}(\alpha, n_j - \alpha)$, with $n_j = n_1 + \sum_{i<j} m_i$ fixed by the step. Here ratings are $\text{Bin}(m_j, C)$, so that beliefs stay in the family.

Both families are ordered by the likelihood ratio, and $\mathbb{E}_b C$ is strictly increasing along them. With MLRP ratings:

(i) the posterior $T_j(b, y)$ is nondecreasing in $b$ and $y$;
(ii) the predictive distribution of $y$ is increasing in $b$ in the first-order stochastic sense;
(iii) hence $b \mapsto \mathbb{E}[\phi(T_j(b, y)) \mid b]$ is nondecreasing for every nondecreasing $\phi$ on $F_{j+1}$.

**Definition 5.** The *investment value* of step $j$ is

$$I_j(b) = \mathbb{E}[V_{\text{end}}(T_j(b, y)) \mid b] - V_{\text{end}}(b) = \tilde{\kappa}\Big(\mathbb{E}\big[\Psi(T_j(b, y))^+ \mid b\big] - \Psi(b)^+\Big) \geq 0.$$

The second expression follows from Lemma 2 and the martingale property; the sign follows from convexity.

**Theorem 6** (Threshold rules)**.** *Assume S1–S4 with $q$ known, and assume for every $j$:*

$$\textbf{(M)} \qquad I_j(b_2) - I_j(b_1) \;\geq\; -v_j\big(\mathbb{E}_{b_2}C - \mathbb{E}_{b_1}C\big) \qquad \text{for all } b_1 \leq b_2 \text{ in } F_j.$$

*Then for every step $j$ and budget $B$, $G_j(\cdot, B)$ is nondecreasing on $F_j$. The set of beliefs at which the agent asks is therefore $\{b \in F_j : \mathbb{E}_b C \geq c_j^*(B)\}$, a threshold on $\mathbb{E}[C]$, possibly at 0 or above 1.*

*Proof.* Let $O_j = V_j - V_{\text{end}} = \hat{p}_j^{+} G_j^{+}$, and note $O_{J+1} \equiv 0$. Suppose $O_{j+1}(\cdot, B')$ is nondecreasing on $F_{j+1}$ for every $B'$; this holds for $j = J$. Since $V_{\text{end}}$ does not depend on $B$,

$$G_j(b, B) = \sup_{s \in S_j(B)} \Big\{ W_j(b; s) - c_j^{\text{time}} + \mathbb{E}\big[O_{j+1}(T_j(b, y), B - \sigma(k_j + s)) \mid b\big] \Big\} + I_j(b).$$

Fix $b_1 \leq b_2$ and $\varepsilon > 0$, and let $s_1$ be $\varepsilon$-optimal at $b_1$. Then:

- $W_j(b_2; s_1) - W_j(b_1; s_1) = v_j\big(1 + g(s_1)\big)(\mathbb{E}_{b_2}C - \mathbb{E}_{b_1}C) \geq v_j(\mathbb{E}_{b_2}C - \mathbb{E}_{b_1}C)$, because $g \geq 0$;
- the $O_{j+1}$ term is nondecreasing in $b$, by (iii) and the induction hypothesis;
- (M) bounds the fall in $I_j$.

Hence $G_j(b_2, B) - G_j(b_1, B) \geq -\varepsilon$ for every $\varepsilon > 0$. So $G_j(\cdot, B)$ is nondecreasing, and then so is $O_j(\cdot, B) = \hat{p}_j^{+} G_j^{+}$. Induction completes the proof. □

Condition (M) asks the consumption value of the step to rise with $\mathbb{E}[C]$ at least as fast as the step's investment value can fall. The investment value is the value of what the step reveals for the after-date choice. The next corollary gives conditions on primitives.

**Corollary 7.** *Condition (M), and so the conclusion of Theorem 6, holds in each of the following cases.*

*(a)* The after-date choice does not depend on what the date reveals: $\tilde{\kappa} = 0$*, or $\psi$ has constant sign on the support of beliefs. Then $I_j \equiv 0$.*

*(b)* Two types: $v_j(C_H - C_L) \geq \tilde{\kappa}\,|\psi(C_H) - \psi(C_L)|$ *for every $j$. This does not depend on the rating technology.*

*(c)* Beta beliefs: $\Psi$ *is nondecreasing on every $F_j$, and $\tilde{\kappa} L_j \leq v_j$, where*

$$L_j = \sup_{b_1 < b_2 \in F_j} \frac{\Psi(b_2) - \Psi(b_1)}{\mathbb{E}_{b_2}C - \mathbb{E}_{b_1}C}.$$

*If $\psi$ is nondecreasing on $[0, 1]$, both parts hold whenever $\tilde{\kappa} \sup_c \psi'(c) \leq \min_j v_j$.*

*Proof.* (a) If $\psi \geq 0$ on the support, $\Psi \geq 0$ at every posterior, so $I_j = \tilde{\kappa}(\mathbb{E}[\Psi(b')] - \Psi(b)) = 0$ by (2). The case $\psi \leq 0$ is the same, and $\tilde{\kappa} = 0$ is immediate.

(b) Write $x_0 = \psi(C_L)$, $x_1 = \psi(C_H)$ and $P_y^H = P_j(y \mid C_H)$, $P_y^L = P_j(y \mid C_L)$. Then

$$I_j(\pi)/\tilde{\kappa} = \sum_y \big(\pi P_y^H x_1 + (1 - \pi) P_y^L x_0\big)^{+} - \big(x_0 + \pi(x_1 - x_0)\big)^{+}.$$

This is continuous and piecewise linear in $\pi$. Where it is differentiable, its slope is

$$\sum_{y \in A(\pi)} \big(P_y^H x_1 - P_y^L x_0\big) - \mathbf{1}\{\Psi(\pi) > 0\}(x_1 - x_0),$$

where $A(\pi)$ is the set of $y$ whose term is positive.

- If $x_0 < 0 < x_1$, every summand is positive, so the slope is at least $-(x_1 - x_0)$.
- If $x_1 < 0 < x_0$, every summand is negative, so the sum is at least $\sum_y (P_y^H x_1 - P_y^L x_0) = x_1 - x_0$, and the indicator term is non-negative.
- If $x_0$ and $x_1$ have the same sign, (a) applies.

So the slope of $I_j$ in $\pi$ is at least $-\tilde{\kappa}|x_1 - x_0|$. Since $\mathbb{E}_\pi C = C_L + \pi(C_H - C_L)$, (M) holds under the stated inequality.

(c) For $b_1 \le b_2$ in $F_j$, write

$$\frac{I_j(b_2) - I_j(b_1)}{\tilde{\kappa}} = \Big[\mathbb{E}\big(\Psi^+(b') \mid b_2\big) - \mathbb{E}\big(\Psi^+(b') \mid b_1\big)\Big] - \Big[\Psi^+(b_2) - \Psi^+(b_1)\Big].$$

The first bracket is non-negative by (iii), because $\Psi^+$ is nondecreasing on $F_{j+1}$. The second is at most $(\Psi(b_2) - \Psi(b_1))^+ \le L_j(\mathbb{E}_{b_2} C - \mathbb{E}_{b_1} C)$. If $\psi$ is nondecreasing, first-order dominance gives

$$\Psi(b_2) - \Psi(b_1) = \int \psi'(c)\,\big(F_{b_1}(c) - F_{b_2}(c)\big)\,dc \le \sup \psi' \cdot \big(\mathbb{E}_{b_2} C - \mathbb{E}_{b_1} C\big),$$

and $\Psi$ is nondecreasing. □

**Proposition 8** (Conjecture 1 fails without (M))**.** *Take two types $C \in \{0.2, 0.8\}$ and one step left, with $v = 50$, $k = 30$, $c^{\text{time}} = 0$, $\sigma = \lambda = 1$ and one Bernoulli rating ($m = 1$). Let $q = 0.9$, D1 with $\eta = 0.2$, $q_D = 0.8$, $\bar{h} = 0.5$, $\delta_D = 0.9$, $R = 6000$, $U = 1000$. Then the agent asks at the last step on $[0.0706,\ 0.2311] \cup [0.6667,\ 1]$ and ends on the rest of $[0, 1]$. The set of asking beliefs is not an interval, and no threshold rule is optimal.*

*Proof.* The primitives give:

- $\psi(0.2) = -78.75$ and $\psi(0.8) = 676.19$, so $\Psi(\pi) = 0$ at $\pi_k = 0.1043$;
- $\tilde{\kappa} = 0.8 - 0.2 \cdot 0.2 = 0.76$ and $\hat{p} = 0.9 - 0.2 \cdot 0.1 > 0$;
- $s^* = \max(50\,\mathbb{E}C - 30, 0)$, which is 0 for $\mathbb{E}C \le 0.6$, so $W(\pi) = 50\,\mathbb{E}_\pi C - 30$ there.

Consider four regions.

- For $\pi \ge 0.318$, the posterior after a low rating, $\pi/(\pi + 4(1 - \pi))$, is above $\pi_k$. So $\Psi^+$ is linear on the support of the posteriors, and $I(\pi) = 0$. Then $G = W < 0$ for $\mathbb{E}_\pi C < 0.6$, that is, for $\pi < 0.667$; and $G \ge 0$ above.

- At $\pi = \pi_k$, only a high rating leads to asking for another date. So

  $$I(\pi_k) = \tilde{\kappa}\big(\pi_k \cdot 0.8 \cdot 676.19 + (1 - \pi_k) \cdot 0.2 \cdot (-78.75)\big) = 32.17,$$

  while $W(\pi_k) = -16.87$, giving $G(\pi_k) = 15.29 > 0$.

- At $\pi = 0.05$, $G = -9.32 < 0$.

$G$ therefore changes sign three times. `verify.py` computes $G$ on a fine grid and locates the switch points by bisection. □

The mechanism is the after-date option. Near $\pi_k$ the agent is close to indifferent between cutting contact and asking for a second date. The last step's rating settles that choice, so the step is worth taking for what it reveals, even though it is not worth its price as an evening out. At somewhat higher beliefs the second date is already decided, the information is worthless, and the agent ends. Here (M) fails badly: $v(C_H - C_L) = 30$, while $\tilde{\kappa}|\psi(C_H) - \psi(C_L)| = 574$.

*Remark* 9. In this example, where (M) fails, the asking set is a union of two intervals in $\mathbb{E}[C]$. A card can record it as such, with extra columns near the after-date indifference point. We do not prove a general bound on the number of intervals.

# 3 Conjectures 2 and 3: the outside option and the value of a relationship

**Scope.** Conjectures 2 and 3 are stated in Appendix A.1, that is, in M2+. Conjecture 3 sits under "the full recursion", which replaces the truncation (21). The results of this section are for the core model with the truncated $V_{\text{end}}$. Their proofs use $\ell = 0$ (Lemma 3), which fails in the full recursion. They therefore settle the core-model analogues of Conjectures 2 and 3, and show that the conjectured signs are not implied by the structure the core model shares with M2+. The M2+ statements themselves remain open.

$U$ and $R$ enter the core model only through $V_{\text{end}}$, that is, through $a$ and $\psi$. One lemma covers both.

**Lemma 10** (Pivotality)**.** *Assume S1–S4. Beliefs about $q$ and the budget are arbitrary. Let a parameter change from $\theta$ to $\theta'$ change $a$ arbitrarily and raise $\psi$ pointwise, $\psi_{\theta'} \geq \psi_\theta$ on $[0,1]$, leaving everything else unchanged. Let*

$$D_j = V_j^{\text{ask}} - V_{\text{end}} = \hat{p}_j G_j$$

*be the value of asking minus the value of ending at step $j$ in the fictitious problem of Section 1. When $\hat{p}_j > 0$, the agent asks iff $D_j \geq 0$; the real gain from asking has the same sign.*

*(i) If $\Psi_\theta(b) > 0$, so that the agent would ask for another date, then $D_j(b;\theta') \leq D_j(b;\theta)$.*

*(ii) If $\Psi_{\theta'}(b) \leq 0$, so that the agent would cut contact, then $D_j(b;\theta') \geq D_j(b;\theta)$.*

*Proof.* In the fictitious problem, the value of any policy $\pi$ that asks now is

$$\Phi(\pi;\theta) = \tilde{\mathbb{E}}^\pi\Big[\sum \text{flows}\Big] + \tilde{\mathbb{E}}^\pi\big[V_{\text{end}}(b_\tau;\theta)\big],$$

and $V_j^{\text{ask}} = \sup_\pi \Phi(\pi;\theta)$. Neither the flows nor $\tilde{\mathbb{P}}^\pi$ depend on $\theta$; only the terminal payoff does. Let $\pi$ and $\pi'$ be $\varepsilon$-optimal asking policies at $\theta$ and $\theta'$, and write $\delta V(x) = V_{\text{end}}(x;\theta') - V_{\text{end}}(x;\theta)$. Then

$$\tilde{\mathbb{E}}^\pi[\delta V(b_\tau)] - \varepsilon \;\leq\; V_j^{\text{ask}}(\theta') - V_j^{\text{ask}}(\theta) \;\leq\; \tilde{\mathbb{E}}^{\pi'}[\delta V(b_\tau)] + \varepsilon.$$

By Lemma 2, $\delta V(x) = \mathbb{E}_x[a_{\theta'} - a_\theta] + \tilde{\kappa}\,\chi(x)$, where

$$\chi(x) = \Psi_{\theta'}(x)^+ - \Psi_\theta(x)^+, \qquad 0 \leq \chi(x) \leq \mathbb{E}_x\zeta, \qquad \zeta = \psi_{\theta'} - \psi_\theta \geq 0.$$

The first term of $\delta V$ is linear in $x$, so by (2) it cancels against the same term in $\delta V(b)$.

(i) Here $\Psi_{\theta'}(b) \geq \Psi_\theta(b) > 0$, so $\chi(b) = \mathbb{E}_b\zeta$. Then

$$D_j(\theta') - D_j(\theta) \leq \tilde{\kappa}\big(\tilde{\mathbb{E}}^{\pi'}[\mathbb{E}_{b_\tau}\zeta] - \mathbb{E}_b\zeta\big) + \varepsilon = \varepsilon,$$

using (2) again.

(ii) Here $\Psi_\theta(b) \leq \Psi_{\theta'}(b) \leq 0$, so $\chi(b) = 0$. Then

$$D_j(\theta') - D_j(\theta) \geq \tilde{\kappa}\,\tilde{\mathbb{E}}^\pi[\chi(b_\tau)] - \varepsilon \geq -\varepsilon.$$

Let $\varepsilon \to 0$. □

The intuition is that the direct value of $V_{\text{end}}$ is linear in beliefs, so it is the same, in expectation, whether the agent continues or ends. A parameter therefore changes the value of continuing only through the information value of the date for the after-date choice. That value rises when the parameter moves the after-date choice towards indifference, and falls when it moves the choice away.

**Theorem 11** (Conjecture 2, core-model analogue)**.** *Assume S1 and S3, and let $U < U'$.*

*(a)* Phase C (true). *The set of beliefs at which the agent asks for another date shrinks. The agent asks iff $\hat{p}(P_D) > 0$ and $V_D(b_{\text{yes}}) > U$, and*

$$V_D - U = \delta_D \mathbb{E}[hC]R - \big(1 - \delta_D(1 - \mathbb{E}h)\big)U$$

*is strictly decreasing in $U$. This part does not need S4.*

*(b)* On the day (true only where the agent would cut contact). *Assume S4 and let $A_j(U)$ be the set of beliefs at which the agent asks at step $j$.*

- *On $\{b : \Psi_U(b) \le 0\}$, where the agent would cut contact after the date: $A_j(U') \subseteq A_j(U)$. Raising $U$ makes continuing harder, as conjectured.*
- *On $\{b : \Psi_{U'}(b) > 0\}$, where the agent would ask for another date even at $U'$: $A_j(U) \subseteq A_j(U')$. Raising $U$ makes continuing* easier*, contrary to the conjecture.*

*(c)* Spend. *While the budget does not bind, discretionary spend does not depend on $U$, by Proposition 1.*

*Proof.* (b) We have $\partial\psi/\partial U = -(1-h)\big(1 - \delta_D(1-h)\big) \le 0$, so $\psi_U \ge \psi_{U'}$. Apply Lemma 10 with $\theta = U'$ and $\theta' = U$, recalling that the agent asks iff $\hat{p}_j > 0$ and $D_j \ge 0$, and that $\hat{p}_j$ does not depend on $U$. □

**Example 12** (Conjecture 2's analogue fails on the day, inside a threshold model)**.** Use the model of Proposition 8 with $R = 600$, $U = 100$ and a last step with $v = 100$ and $k = 32$. Corollary 7(b) holds, since $60 \ge 57.4$, so the asking set is a threshold. It still holds after raising $U$ by 5, since $60 \ge 57.0$.

- The threshold is $\pi^* = 0.1605$, where $\Psi = 4.24 > 0$.
- Raising $U$ by 5 *lowers* the threshold to 0.1507.
- With $k = 25$ instead, the threshold is 0.0606, where $\Psi = -3.30 < 0$. There raising $U$ by 5 raises it to 0.0622, as conjectured.

**Theorem 13** (Conjecture 3, first part, core-model analogue)**.** *Assume S1 and S3, and let $R < R'$.*

*(a)* Phase C (true). *The set of beliefs at which the agent asks for another date grows, because $V_D - U$ is increasing in $R$.*

*(b)* On the day. *Assume S4.*

- *On $\{b : \Psi_R(b) > 0\}$: $A_j(R') \subseteq A_j(R)$. A more valuable relationship makes continuing* harder*, contrary to the conjecture.*
- *On $\{b : \Psi_{R'}(b) \le 0\}$: $A_j(R) \subseteq A_j(R')$, as conjectured.*

*In Example 12, raising $R$ by 30 raises the threshold from 0.1605 to 0.1676. With $k = 25$ it lowers the threshold from 0.0606 to 0.0584. At $R = 630$ the sufficient condition of Corollary 7(b) narrowly fails ($60 < 60.65$). That $G$ still crosses zero once is checked numerically in* `verify.py`*.*

*Proof.* $\partial\psi/\partial R = (1-h)\,\delta_D\,h\,c \ge 0$. Apply Lemma 10 with $\theta = R$ and $\theta' = R'$. □

*Remark* 14 (Why $R$ does not simply favour escalation)*.* In the core model the chance of a relationship, $h(C, q_D)$, does not depend on how far up the ladder the date went. The direct value of a relationship, $\mathbb{E}[hC]R$, is therefore a martingale term that continuing cannot raise. If the intended reading is that escalating raises the chance of a relationship, $h$ must depend on the step reached. Conjecture 3 would then need a separate argument.

**Proposition 15** (Conjecture 3, second part: investment value)**.** *Assume S1–S4.*

*(i)* $I_j \geq 0$.

*(ii)* Two types, *with* $x_0 = \psi(C_L) < 0 < x_1 = \psi(C_H)$: $I_j$ *is nondecreasing on* $[0, \pi_k]$ *and nonincreasing on* $[\pi_k, 1]$, *where* $\pi_k = -x_0/(x_1 - x_0)$ *is the after-date indifference point. Moreover* $I_j(\pi) = 0$ *whenever every posterior lies on the same side of* $\pi_k$ *as* $\pi$. *If* $x_0, x_1$ *have the same sign,* $I_j \equiv 0$.

*(iii)* Settling: *for Beta beliefs with* $\mathrm{Bin}(m_j, C)$ *ratings, writing* $\psi(c) = \sum_r \psi_r c^r$, *which has degree at most* 3,

$$I_j\big(\mathrm{Beta}(\alpha, n-\alpha)\big) \leq \tilde{\kappa}\, \frac{m_j}{n} \sum_{r \leq 3} r|\psi_r| \;\longrightarrow\; 0 \qquad (n \to \infty).$$

*(iv)* Across beliefs, investment value is not largest where uncertainty is largest. *With two types,* $\mathrm{Var}(C) = \pi(1-\pi)(C_H - C_L)^2$ *is largest at* $\pi = 1/2$. *In the model of Proposition 8,* $I(1/2) = 0$ *but* $I(\pi_k) = 32.2$, *at* $\pi_k \approx 0.10$, *where the agent is fairly sure the match is incompatible. So any share of the step's value attributed to investment is zero at maximal uncertainty and positive at* $\pi_k$, *however the share is normalised.*

*(v)* At a fixed mean, investment value is not monotone in precision. *With Beta beliefs of mean* 0.3, $I = 16.24, 8.31, 4.69, 4.14, 4.72, 3.56, 0.45$ *for* $n = 1, 2, 4, \ldots, 64$. *In this example the least settled belief,* $n = 1$, *does have the largest value, but* $I$ *rises from* $n = 8$ *to* $n = 16$.

*Proof.* (ii) On $\{\Psi \leq 0\}$, $I/\tilde{\kappa} = \sum_y (\cdot)^+$ with every summand nondecreasing, because its slope $P_y^H x_1 - P_y^L x_0$ is positive. On $\{\Psi \geq 0\}$, $I/\tilde{\kappa} = \sum_y (\cdot)^+ - \sum_y (\cdot) = \sum_y (\cdot)^-$, with every summand nonincreasing. If all posteriors lie on one side of $\pi_k$, $(\cdot)^+$ is linear on them and (2) gives $I = 0$.

(iii) Since $x \mapsto x^+$ is 1-Lipschitz, $I_j \leq \tilde{\kappa}\, \mathbb{E}|\Psi(b') - \Psi(b)|$. For $b' = \mathrm{Beta}(\alpha + y, n - \alpha + m - y)$, each factor of $\mathbb{E}_{b'} C^r = \prod_{i<r} \frac{\alpha+y+i}{n+m+i}$ differs from the corresponding factor of $\mathbb{E}_b C^r$ by at most $m/(n+m+i) \leq m/n$. All factors lie in $[0,1]$, so $|\mathbb{E}_{b'} C^r - \mathbb{E}_b C^r| \leq rm/n$.

(iv) From $\pi = 1/2$ the posteriors are 0.2 and 0.8, both above $\pi_k$, so (ii) gives $I = 0$.

(v) The values are computed in `verify.py`. □

This is the classical point that the value of information is not monotone in prior uncertainty [1, 2]. The verdict on "largest when beliefs are least settled" depends on its reading. Compared across beliefs, it is false: the investment value peaks near the after-date indifference point. Compared at a fixed mean, it is not monotone, but the example does not refute "largest at the least settled belief". Under either reading, the investment value vanishes as beliefs settle.

## 4 Conjecture 4: tight budgets

**Proposition 16.** *Assume S1 and S3.*

*(i)* Shadow value (true, by definition). *If* $V(b, \cdot)$ *is concave,* $\mu(B) = [V(B) - V(B - \varepsilon)]/\varepsilon$ *is nonincreasing in* $B$.

*(ii)* The premise usually fails. *Suppose step* $j$ *has entry price* $k_j > 0$, $\hat{p}_j > 0$, *and* $G_j(b, \sigma k_j) > 0$. *Then* $V_j(b, \cdot)$ *jumps up at* $B = \sigma k_j$ *and is not concave on any interval containing* $\sigma k_j$ *in its interior.*

*(iii)* Thresholds (true in the core model, with no concavity needed). $G_j(b, \cdot)$ *is nondecreasing in* $B$ *for every* $b$ *and* $j$. *So the set of beliefs at which the agent asks shrinks as the budget falls.*

*(iv)* Spend (true under concavity). *If* $V_{j+1}(b', \cdot)$ *is concave for every* $b'$, *the set of optimal spends* $s_j^*(b, B)$ *is ascending in* $B$ *in the strong set order. In particular its greatest and least elements are nondecreasing in* $B$: *with less budget, the agent spends less, as if money*

*were more expensive. By (ii), the premise holds on the whole budget range essentially only when no later step is ever asked. In the core model this mainly covers the last step, where* $V_{J+1} = V_{\text{end}}$ *and* $s^*_J = \min\{s^0, B/\sigma - k_J\}$ *directly.*

*(v)* Cheaper opening activities: not implied by the shadow-value logic, and false without the premise. *The cost of the optimal opening activity need not fall with the price of money (Example 18). Without the premise, it need not rise with the budget (Example 17). We give no counterexample in which* $V$ *is concave in* $B$*.*

*(vi)* Beyond the core, last step (true under the premise). *Suppose the value after the date uses the budget,* $V_{\text{end}}(b, B)$*, as in M2+ where another date must be paid for. If* $V_{\text{end}}(b, \cdot)$ *is concave, then* $G_J(b, \cdot)$ *is nondecreasing in* $B$*.*

*(vii)* Beyond the core, without the premise (false). *If* $V_{\text{end}}(b, \cdot)$ *is not concave, thresholds can* fall *as the budget falls (Example 20).*

*Proof.* (i) Slopes of chords of a concave function are nonincreasing.

(ii) For $B < \sigma k_j$ the step is unaffordable and $V_j = V_{\text{end}}$. At $B = \sigma k_j$, $V_j = V_{\text{end}} + \hat{p}_j G_j > V_{\text{end}}$. A nondecreasing function with an upward jump is not concave across it.

(iii) By induction, $V_{j+1}(b', \cdot)$ is nondecreasing: $V_{J+1} = V_{\text{end}}$ does not depend on $B$, and (1) preserves monotonicity. The feasible set $S_j(B)$ grows with $B$, and $V_{\text{end}}$ does not depend on $B$.

(iv) For $s < s'$, $F(s', B) - F(s, B)$ consists of a part independent of $B$ plus

$$\mathbb{E}\big[V_{j+1}(x) - V_{j+1}(x + d)\big], \qquad x = B - \sigma(k_j + s'), \quad d = \sigma(s' - s).$$

This is nondecreasing in $B$ by concavity. So $F$ has increasing differences in $(s, B)$, $S_j(B) = [0, B/\sigma - k_j]$ is ascending, and Topkis' theorem [3] applies.

(vi) We have

$$G_J(b, B) = \sup_{s \in S_J(B)} \big\{W_J(b; s) - c_J + V_{\text{end}}(b, B - \sigma(k_J + s))\big\} - V_{\text{end}}(b, B).$$

Let $B < B'$, and let $s$ be $\varepsilon$-optimal at $B$. Then $s$ is feasible at $B'$. With $d = \sigma(k_J + s)$, concavity gives $V_{\text{end}}(B' - d) - V_{\text{end}}(B') \geq V_{\text{end}}(B - d) - V_{\text{end}}(B)$. Hence $G_J(b, B') \geq G_J(b, B) - \varepsilon$. □

**Example 17** (Opening activity and the budget)**.** Let $C$ be known, $q = 1$, $\eta = 0$ and $\sigma = \lambda = 1$. Coffee has $x = \mathbb{E}[C]v = 10$ and $k = 8$; dinner has $x = 60$ and $k = 40$. Spend is optimal subject to the budget. A coffee opening leads to coffee then dinner; a dinner opening leads to dinner only.

- $B = 45$: coffee-first is worth 2.23, because dinner is no longer affordable after coffee. Dinner-first is worth 22.07. The agent opens with dinner.
- $B = 80$: coffee-first is worth 26.56 and dinner-first 24.33. The agent opens with coffee.

The richer agent chooses the cheaper opening. This $V(B)$ is not concave, as (ii) predicts.

**Example 18** (Opening activity and the price of money)**.** Read "as if money were more expensive" as a rise in $\lambda$. Let coffee have $x = 10.8$ and $k = 8$, dinner $x = 63$ and $k = 40$, with $\sigma = 1$. Dinner is worth taking in either ladder for all $\lambda$ near $\lambda_0 = 1.35$. Coffee-first minus dinner-first equals the coffee's own net value $W_c(\lambda)$ (Remark 4). This is 0.408 at $\lambda = 1.30$ and $-0.4$ at $\lambda = 1.40$. Dearer money switches the opening from coffee to dinner.

By the envelope theorem, $\partial V^a / \partial \lambda = -\sigma\, \mathbb{E}^a[\text{outlay}]$. A higher $\lambda$ therefore favours the opening with the lower *expected total outlay*, which need not be the cheaper opening.

*Remark* 19 (Heuristic). Suppose that, under the conjecture's own premise, the overall value $V = \max(V^{\text{coffee}}, V^{\text{dinner}})$ is concave, and that both $V^a$ are differentiable at a point where the optimal opening switches. At a transversal switch, a maximum of two differentiable functions has a convex kink, so the switch can only occur at a point of tangency. This suggests the premise sits uneasily with any systematic move to cheaper openings. It is not a proof: the entry-price jumps of Proposition 16(ii) break differentiability.

**Example 20** (M2+: thresholds can fall with the budget)**.** At the last step the net value is $w = 10$ and the entry price $k = 20$, with $\sigma = 1$. Suppose the after-date option is worth 30 only if at least 40 of budget remains for another date. Then

$$G_J(B) = w + V_{\text{end}}(B - 20) - V_{\text{end}}(B).$$

This is 10 at $B = 30$, where a second date is unaffordable anyway; $-20$ at $B = 50$, where continuing would cost the second date; and 10 at $B = 70$. The agent asks at 30, ends at 50, and asks at 70. Here $V_{\text{end}}(\cdot)$ is a step function and not concave; by Proposition 16(vi), the reversal requires that.

# 5 Conjecture 5: spending to learn

Relax A7 as in Appendix A.7: the rating after step $j$ comes from a family $\{P_m\}$ that is increasing in $m$ in Blackwell's order, with $m = m(s)$ nondecreasing in spend. Integer Binomial $m$ qualifies, as does Gaussian precision. Write

$$f_\mu(s) = \mu v_j g(s) - \sigma\lambda s.$$

Proposition 1's spend $s^0(\mu)$ is the unique maximiser of $f_\mu$ on $[0, \infty)$, which is strictly concave when $\mu v_j > 0$.

**Proposition 21.** *Assume S1–S4, a slack budget and $\hat{p}_j > 0$. Beliefs about $q$ are arbitrary, and the sting model is any of D1, D2, KR.*

*(i)* Optimal spend is never below Proposition 1's spend. *Every optimal spend satisfies $s_j^*(b) \geq s^0(\mathbb{E}_b C)$, under all three sting models.*

*(ii)* It falls back as beliefs settle. *Suppose $m(\cdot)$ takes finitely many values, and beliefs settle: $b_n \to \delta_{\bar{c}}$ weakly, for example Beta beliefs with $n \to \infty$ and mean $\to \bar{c}$. Then $s_j^*(b_n) \to s^0(\bar{c})$.*

*(iii)* "Peaks under uncertainty" is false. *Take two types and the after-date model of Proposition 8, at a last step with $v = 50$ and $k = 20$. Spend buys a binary signal of accuracy*

$$\omega(s) = \tfrac{1}{2} + 0.35(1 - e^{-s/20}) < 1 - \pi_k.$$

*At $\pi = 1/2$, where uncertainty is greatest, the information premium is exactly zero: $s^* = s^0 = 5$. At $\pi_k = 0.104$, $s^* = 20.44$ while $s^0 = 0$. The agent asks at both beliefs: the gain from asking at $s^*$ is 5.58 at $\pi = 1/2$ and 5.95 at $\pi_k$.*

*Proof.* (i) By Lemma 3, the optimal venue maximises $\Phi(s) = f_\mu(s) + e(s)$, with

$$e(s) = \mathbb{E}\big[V_{j+1}(b'_{m(s)}) \mid b\big].$$

$V_{j+1}$ is convex in the belief about $C$ (Theorem 24(a)), so Blackwell's theorem makes $e$ nondecreasing. If $s < s^0$, then $f_\mu(s) < f_\mu(s^0)$ and $e(s) \leq e(s^0)$, so $s$ is not optimal.

(ii) We first show $V_{j+1}$ is continuous in $b$ for the weak topology on beliefs about $C$, including degenerate beliefs. Each ingredient is: $V_{\text{end}}(b) = \mathbb{E}_b a + \tilde{\kappa}\Psi(b)^+$; $W_j$, which depends on $\mathbb{E}_b C$; and the preposterior map $b \mapsto \sum_y P(y \mid b)V(b'_y)$. In the last of these, $P(y \mid b) = \mathbb{E}_b[P(y \mid C)]$ is continuous, the unnormalised posterior is weakly continuous, and $|P(y \mid b)V(b'_y)| \leq P(y \mid b) \sup |V|$ near beliefs with $P(y \mid b) = 0$.

Hence $\text{EVSI}_m(b) = e_m(b) - V_{j+1}(b)$ is continuous, non-negative, and zero at degenerate beliefs. So

$$\varepsilon_n = \max_m \text{EVSI}_m(b_n) \to 0.$$

Maximisers of $\Phi_n = f_{\mu_n} + e$ lie in a fixed compact set, because $f_{\mu_n} \to -\infty$ linearly while $e$ varies by at most $\varepsilon_n$. On that set, $f_{\mu_n} \to f_{\bar{c}}$ uniformly. Since $f_{\bar{c}}$ is strictly concave with unique maximiser $s^0(\bar{c})$, the standard argument applies: for $|s - s^0| \geq \epsilon$, $f_{\bar{c}}(s) \leq f_{\bar{c}}(s^0) - \gamma_\epsilon$. Once $2\varepsilon_n + 2\sup|f_{\mu_n} - f_{\bar{c}}| < \gamma_\epsilon$, every maximiser of $\Phi_n$ is within $\epsilon$ of $s^0$.

(iii) From $\pi = 1/2$ the low posterior is $1 - \omega(s) > \pi_k$, so both posteriors lie above the kink and EVSI $\equiv 0$ for every $s$. At $\pi_k$, $\mathrm{EVSI}(s) = \tilde{\kappa}(2\omega(s) - 1)\,\pi_k\,\psi(C_H)$ is strictly increasing in $s$. The optimal spends and the gains from asking are computed in `verify.py`. □

*Remark* 22. The proposal's caveat that "with psychological costs of refusal EVSI is not guaranteed to be non-negative" does not bite here. The information that spend buys in A.7 is about compatibility, and Theorem 24(a) shows that such information is always weakly valuable in the core model, under all three sting models.

# 6 The value of information (Section 4.4)

**Definition 23.** Fix a belief $b_C$ about $C$. A function $V$ of beliefs about $q$ is in the class $\mathcal{C}$ if it is nondecreasing in the likelihood-ratio order, and convex along every *MLR segment*, that is, along $t \mapsto (1-t)b_1 + tb_2$ for $b_1 \leq_{\mathrm{LR}} b_2$.

**Theorem 24** (Sign of the value of information)**.** *Assume S1–S4.*

*(a)* Information about compatibility. *For every $j$, every $B$ and every belief about $q$, $V_j$ is convex in the belief about $C$. So every signal about $C$ has non-negative EVSI under D1, D2 and KR. This includes ratings and the opening activity's signal in Definition 2.*

*(b)* Information about willingness. *Suppose the sting satisfies*

$$\textbf{(K)} \qquad \begin{array}{c} f(p) = p - \kappa(p) \text{ is convex and twice differentiable,} \\ \text{and } \kappa(p)/p \text{ is nonincreasing on } (0,1]. \end{array}$$

*D1, D2 and KR all satisfy (K). Suppose also that the cues observed during the date are MLRP signals about $q$. Then every MLRP signal about $q$ has non-negative EVSI at every decision point of the date.*

*(c)* Non-monotone information about willingness can hurt. *Under D1, D2 and KR, some signals about $q$ that are not MLR-ordered have strictly negative EVSI (Examples 25 and 26).*

*Proof of (a).* We argue by backward induction using Lemma 3, writing

$$V_j = \max\Big\{V_{\mathrm{end}},\ \hat{p}_j^{+}\big(\sup_s[W_j(b;s) - c_j + \mathbb{E}(V_{j+1}(b') \mid b)]\big) + (1 - \hat{p}_j^{+})V_{\mathrm{end}}\Big\}.$$

Each ingredient is convex in $b_C$:

- $V_{\mathrm{end}}$ is convex (Lemma 2);
- $W_j(b;s)$ is affine in $b_C$;
- $b_C \mapsto \sum_y P(y \mid b_C)V_{j+1}(b'_y)$ is a sum of perspectives of the convex function $V_{j+1}$, hence convex;
- a supremum of convex functions is convex;
- $\hat{p}_j^{+} \in [0,1]$ does not depend on $b_C$, so the middle term is a convex combination of convex functions.

A maximum of convex functions is convex. Convexity of the value in beliefs is equivalent to non-negative EVSI for every signal. □

*Proof of (b).* Three facts about MLR segments are standard. Let $b_t = (1-t)b_1 + tb_2$ with $b_1 \leq_{\mathrm{LR}} b_2$.

- $b_t$ is increasing in $t$ in the likelihood-ratio order.
- For any likelihood $\ell(z \mid q)$, the posterior $b_{t,z}$ lies on the MLR segment from $b_{1,z}$ to $b_{2,z}$, with weight $tP(z \mid b_2)/P(z \mid b_t)$ on $b_{2,z}$.
- In particular, after a yes ($\ell = \varphi q$), the posterior lies on the segment from $b_{1,\text{yes}}$ to $b_{2,\text{yes}}$ at weight $s(t) = t\bar{p}_2/\bar{p}(t)$, where $\bar{p}(t) = \mathbb{E}_{b_t} q$ is affine and nondecreasing in $t$.

*Step 1: expectations over MLRP signals preserve $\mathcal{C}$.* Let $V \in \mathcal{C}$ and let $z$ be an MLRP signal. Along a segment,

$$\mathbb{E}[V(b_{t,z})] = \sum_z \big(A_z(t) + B_z(t)\big)\, \xi_z\Big(\frac{B_z(t)}{A_z(t) + B_z(t)}\Big),$$

with $A_z(t) = (1-t)P(z \mid b_1)$ and $B_z(t) = tP(z \mid b_2)$ affine, and $\xi_z$ convex. This is a perspective, hence convex in $t$. Monotonicity follows from MLRP. Expectations over ratings (independent of $q$) and suprema over spend also preserve $\mathcal{C}$.

*Step 2: the ask step preserves $\mathcal{C}$.* Suppose $X \in \mathcal{C}$, and recall from Lemma 3 that $V_j - V_{\text{end}} = f(p)^+ X(b_{\text{yes}})^+$.

Monotonicity: by (K), $f = p \cdot (f/p)$ is nondecreasing where positive; $p$ is nondecreasing in the order; and $b \mapsto b_{\text{yes}}$ is monotone.

Convexity along a segment: let $g(t) = f(p(t))\, \xi(s(t))$, where $\xi(s) = X$ along the yes-segment is convex and nondecreasing, and $p(t) = \varphi_j \bar{p}(t)$. Since $p'/p = \bar{p}'/\bar{p}$, we have $s' = \bar{p}_1 \bar{p}_2/\bar{p}^2 \geq 0$ and $s'' = -2\bar{p}' s'/\bar{p} = -2p's'/p$. Suppose first that $\xi \in C^2$. Then

$$g'' = f''\, p'^2\, \xi + f\, \xi''\, s'^2 + 2\, p' \xi' s' \Big(f' - \frac{f}{p}\Big).$$

On the set where $f > 0$ and $\xi > 0$, every term is non-negative:

- $f'' \geq 0$ by (K);
- $p' \geq 0$, because LR order implies ordered means;
- $\xi' \geq 0$ and $\xi'' \geq 0$;
- $f' - f/p = p\,(f/p)' \geq 0$ by (K).

For general $\xi$, the set $T = \{t : f(p(t)) > 0,\ \xi(s(t)) > 0\}$ is open in $[0,1]$ and, as shown next, an interval. Fix a compact subinterval $K \subset T$ on which $s(t)$ stays in $[\underline{s}, \bar{s}] \subset (0,1)$. On $[\underline{s}, \bar{s}]$, $\xi$ is a uniform limit of $C^2$ convex nondecreasing functions $\xi_n$ (mollify), and $\xi_n > 0$ on $s(K)$ for $n$ large. So $g_n = f\, \xi_n \circ s$ is convex on $K$, and so is its uniform limit $g$. Convexity on every compact subinterval gives convexity on $T$, and continuity extends it to $\bar{T}$.

Because $f(p(t))$ is nondecreasing where positive and $\xi \circ s$ is nondecreasing, $\bar{T}$ is an interval $[t_0, 1]$. If $t_0 > 0$, then $g(t_0) = 0$ by continuity, since $s(t_0) \in (0,1)$ and $\xi$ is continuous there. So $f^+\xi^+$ is 0 on $[0, t_0]$ and convex nondecreasing on $[t_0, 1]$, hence convex on $[0,1]$.

Induction on $j$, starting from $V_{J+1} - V_{\text{end}} \equiv 0$, gives $V_j(b_C, \cdot) - V_{\text{end}}(b_C) \in \mathcal{C}$.

*Step 3: from convexity on segments to non-negative EVSI.* Let $z \in \{z_1 < \cdots < z_N\}$ be an MLRP signal. Reveal it in stages: first whether $z = z_N$, then, given $z < z_N$, whether $z = z_{N-1}$, and so on. At each stage MLRP implies $b_{\{z = z_i\}} \geq_{\text{LR}} b_{\{z < z_i\}}$, and the current belief lies on the MLR segment between them. So the expected value weakly rises at each stage. The tower property gives $\mathbb{E}[V(b_z)] \geq V(b)$. □

**Example 25** (Negative EVSI under D1)**.** Let $C$ be known, normalise $V_{\text{end}} = 0$, and take two steps left with net values $w_1 = 0$ and $w_2 = 1$. Let $\varphi_2 = 1$, D1 with $\eta = 1/2$, so $f(p) = 1.5p - 0.5$. Let $q \in \{0.2, 0.6, 1\}$ with prior $(0.3, 0.5, 0.2)$. The signal reveals whether $q$ is extreme, $\{0.2, 1\}$, or middling, $\{0.6\}$. It is not MLR-ordered. The value is

$$V(b) = f(\mathbb{E}_b q)\big(w_1 + f(\mathbb{E}_b[q \mid \text{yes}])\, w_2\big).$$

- Prior: $\mathbb{E}q = 0.56$ and $\mathbb{E}[q \mid \text{yes}] = 0.7$, so $V = 0.34 \times 0.55 = 0.1870$.
- Posterior "extreme", probability $1/2$, $b_1 = (0.6, 0, 0.4)$: $V = 0.28 \times 0.7231 = 0.2025$.
- Posterior "middling", $b_2 = \delta_{0.6}$: $V = 0.4 \times 0.4 = 0.16$.

So EVSI $= 0.1812 - 0.1870 = -0.0058 < 0$. With $\eta = 0$ the EVSI is exactly zero.

The signal moves the chance of a yes now, $\mathbb{E}q$, and the chance of a yes later, $\mathbb{E}[q \mid \text{yes}]$, in opposite directions. That breaks the third term of $g''$ in the proof of (b).

**Example 26** (Negative EVSI under D2 and KR)**.** Let $q \in \{0.1, 0.8, 1\}$ with prior $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, and use the same "extreme versus middling" signal. Take D2 with $\eta = 2$, equivalently KR with $\eta(\Lambda - 1) = 2$, so $f(p) = p(2p - 1)$; and take $w_1 = -\frac{1}{2}$, $w_2 = 1$, $\varphi_2 = 1$.

- Prior: $V = 0.2363 \times 0.0906 = 0.0214$.
- Posterior "extreme": $V = 0.055 \times 0.2679 = 0.0147$.
- Posterior "middling": $V = 0$. Step 2 would still be asked for ($\hat{p} = 0.48 > 0$), but its value 0.48 no longer covers step 1's net loss: $G_1 = -0.5 + 0.48 = -0.02 < 0$.

So EVSI $= 0.0074 - 0.0214 = -0.0140 < 0$.

**Answer to the open question.** In the core model with $q_D$ known, consider signals informative about $C$ alone or about $q$ alone. Among these, only information about willingness that is not ordered by the likelihood ratio can have negative value. Ratings, and cues of the kind used in Section 5.4, which are MLRP, always have non-negative value. This is consistent with the proposal's numerical search, which found no counterexample. Three cases are not covered: signals informative about both $C$ and $q$, signals about $q_D$ when it is uncertain, and settings with $\ell > 0$, such as the full recursion of M2+ or the joint prior of A.3.

# 7 The conjectural prediction about opening activities

Prediction 6 of Appendix A.9 says that more uncertainty before the date favours a more informative opening activity, depending on where beliefs sit relative to the decision threshold. As stated, it is conditional and not precise enough to prove or refute. What the results above give is:

- the informativeness premium of a Blackwell-more-informative opening is never negative (Theorem 24(a) and Blackwell's theorem);
- the premium vanishes as beliefs settle (the argument of Proposition 21(ii));
- for the part of the premium that comes from the after-date choice, with two types, the premium is largest near the after-date indifference point and can be zero at maximal uncertainty (Proposition 15). The full opening premium also involves the on-the-day kinks of $V_1$, and we do not locate its peak.

# 8 Suggested changes to the proposal

1. Replace Conjecture 1 by Theorem 6 with Corollary 7, and cite Proposition 8. Where (M) fails, the card may need more than one boundary near the after-date indifference point.

2. For the core model, add Lemma 10 and Theorems 11–13. Keep the Phase C statements, which are true. Note that in the core model $R$ affects escalation only through information, because $h$ does not depend on the step reached. Keep Conjectures 2 and 3 as open M2+ statements, noting that the core model shows their signs are not structural. Prediction 2 (market thickness) and Prediction 5 (relationship value) of Appendix A.9 should carry the same caveat.

3. Replace "largest when beliefs are least settled", in Conjectures 3 and 5 and in Prediction 6, by "largest near the after-date indifference point, and vanishing as beliefs settle".

4. In Conjecture 4, note that the concavity premise fails whenever entry prices bind. State Proposition 16(iii), (iv) and (vi) instead. Weaken "cheaper opening activities", in the conjecture and in Prediction 3, given Examples 17 and 18.

5. In Section 4.4 and the "Convexity" bullet of Section 8, replace the open question with Theorem 24.

6. Technical points:
   - state the convention $\kappa\Delta^{+}$;
   - note that the fractional likelihood is a quasi-likelihood, so results relying on the martingale property are exact only with integer $m_a$;
   - record the closed form in Remark 4;
   - record, extending the proposal's own Remark after Proposition 3, that $\ell = 0$ at every step of the core model (Lemma 3), so the decision to ask on the day reduces to $p \geq p^{*}_{m}(0)$ together with $G \geq 0$.

## Numerical verification

`python3 verify.py` reproduces every number above and runs randomised checks. Its labels refer to results by name, not number. It checks:

- Corollary 7(b)–(c): monotone $G_j$ whenever the condition holds;
- the sign pattern of Lemma 10 for $U$ and $R$, at several thousand belief points;
- Proposition 16(iii);
- Theorem 24(a), as convexity of $V_0$ in beliefs about $C$;
- Theorem 24(b), as non-negative EVSI of Binomial cues about $q$ under general priors.

The output is saved in `verify_output.txt`.

## References


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